RAMAIAH
Institute of Technology

Dept. of Computer Science and Engineering (A
& Computer Science and Engineering (Cyber Secu

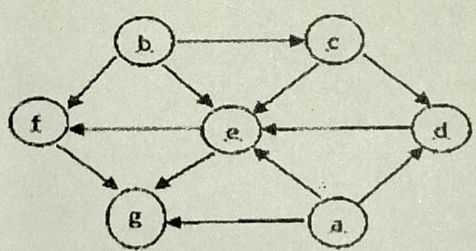
Programme: B. E. – Computer Science and Engineering (AI&ML)
Computer Science and Engineering (Cyber Security)
Internal Assessment – I

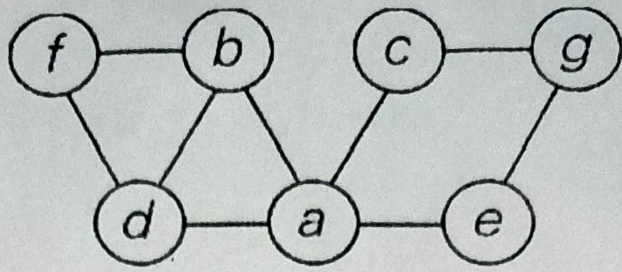
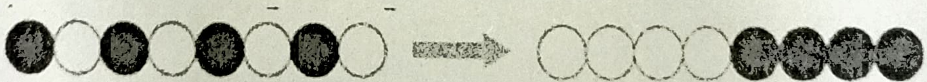
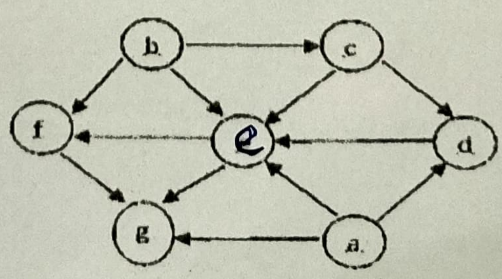
TERM: 3rd March 2025 to 21st June 2025	**COURSE NAME:** Design and Analysis of Algorithms
DATE : 25/04/2025 **TIME**: 11:00 AM to 12:00 PM	**COURSE CODE** : CI43/CY43
MAX MARKS: 30	**PORTIONS** : L1 to L18

Mobile Phones are banned

Instructions to Candidates: **Answer any two full questions.**

Marks: 1

Q. NO	**Questions**	Blooms Levels (L1 to L6)*	CO
1.a	Discuss how asymptotic bounds are used to represent the performance of algorithms and explain the properties involved in measuring their efficiency.	L2	CO1 4
1.b	Illustrate the tracing of the Quick sort algorithm for the given array of numbers and compute the average time complexity. 2, 7, 8, 3, 1, 9, 5, 6, 3	L3	CO2 5
1.c	Consider the graph given in Figure 1(c)  Fig 1(c) Graph i. Starting at vertex 'a' and resolving ties by the vertex alphabetical order, traverse the graph by Breadth First Search (BFS). Illustrate the queue operations for performing BFS Traversal to obtain BFS Spanning Tree. ii. Specify all possible BFS Traversal sequences.	L3	6 CO2
2.a	Design an algorithm to search an element in an array using linear search 10,5,6,3,1,2,7. Derive the best case, worst case and average case efficiency of this algorithm.	L3	5 CO

2.b	Consider the graph given in Figure 2(b)  Fig 2(b) Graph i. List the adjacency matrix and adjacency list for the graph in 2.b, starting from vertex ‘a’. ii. Starting at vertex ‘a’ and resolving ties by the vertex alphabetical order, traverse the graph by Depth First Search (DFS). Illustrate the stack operations for performing DFS Traversal to obtain DFS Spanning Tree.	L3	5 CO2
2.c	Apply the master theorem and find the time complexity for the following equations. i) $T(n) = 3T\left(\frac{n}{2}\right) + n^4$ ii) $T(n) = 4T\left(\frac{n}{2}\right) + n^3$ iii) $T(n) = T\left(\frac{n}{2}\right) + 2^n$	L5	5 CO2
3.a	Construct a heap for the list 1, 8, 6, 5, 3, 7, 4 by the bottom-up approach. by showing every step of heap construction to obtain heap sort. a) Building Max/Min Heap b) Maximum/Minimum Key Deletion from a heap.	L3	5 CO3
3.b	Design an algorithm for the given scenario. You are given a list of 2n disks, n dark and n light, **in a random order**. Each disk is labeled either 'D' for dark or 'L' for light. Your goal is to **sort the disks** so that **all light disks come before all dark disks**, using any sorting algorithm. Disks can be moved around freely in memory (like in arrays). 	L4	5 CO2
3.c	Apply the source elimination graph to represent vertices in topological order for the graph given below and provide the sort order. 	L3	5 CO2